

Qhelile Ozias Sibanda

PhD Student, Computer Science • University of California, Merced

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Research Interests	Personal	Education	Research Experience
Teaching Experience	Publications	Academic Services	Affiliations
Research Projects	Personal Projects	Posters & Presentations	Professional Experience
Skills	Awards	Hobbies	References

RESEARCH INTERESTS

Human-Computer Interaction (HCI), Virtual and Augmented Reality, Accessibility, Applied Machine Learning

PERSONAL

Nationality: Zimbabwean

Languages: English, Ndebele, Zulu and Shona

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EDUCATION

2023 - CURRENT	PhD Student in Computer Science <i>University of California, Merced</i> - CGPA: 3.75/4.0 - Advisor: Dr. Ahmed Arif - Research focus: Novel input and interaction techniques for immersive virtual environments
2021	B.Sc. Honors in Computer Science (First Class) <i>National University of Science and Technology, Zimbabwe</i> CGPA: 3.60/4.0
2016	3 A' Level passes <i>Mathematics, Physics, Geography</i>
2014	12 O' Level passes <i>Including English and Mathematics</i>

RESEARCH EXPERIENCE

2023 - CURRENT	Graduate Student Researcher <i>Inclusive Interaction Lab (IIL), University of California, Merced</i> - Conduct HCI research on virtual reality (VR) input and interaction techniques and on telepresence robot control using the Ohmni robot platform. - Design and run user studies comparing gesture-based, smartwatch, and VR-based control methods for robot teleoperation. - Perform quantitative analysis of user performance and preference data across different
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interaction techniques to inform interface design.

- Visualize findings with clear reports and presentations for the Inclusive Interaction Lab research community.

SPRING 2025

Relational Database Implementation

University of California, Merced • C++, SQLite, Bison/Yacc, Flex, Valgrind, GDB, Linux, GitHub

- Designed and implemented a full-stack, single-user relational database system from scratch across five cumulative project phases, covering the complete SQL query processing pipeline.
- Stage 1 - Catalog: built a catalog system maintaining metadata about tables and attributes, integrated into a persistent SQLite backend.
- Stage 2 - Query Compiler: developed a compiler transforming parsed SQL queries into executable relational algebra trees using lexer/parser tools (lex/yacc).
- Stage 3 - Single-Table Operators: implemented core relational algebra operators including Scan, Select, Project, Duplicate Removal, Sum, and Group By.
- Stage 4 - Join Operators: developed three in-memory join algorithms: Nested Loop Join, Hash Join, and Symmetric Hash Join.
- Stage 5 - Complete Database Server: integrated all components into a command-line SQL engine with B+ Tree indexing and index-based scans (ScanIndex).

FALL 2024

DynaQ Reinforcement Learning Simulation in Custom Grid World

University of California, Merced

- Developed a complete simulation framework implementing the Dyna-Q reinforcement learning algorithm using a custom dungeon-like grid environment (DungeonEnvironment).
- Integrated a Trivial Model to simulate experience replay by learning from both real and hallucinated experiences.
- Implemented epsilon-greedy exploration, Q-learning updates, and parameter tuning (epsilon, alpha, gamma, planning steps) to optimize agent learning behaviour.
- Visualized the agent's learned policy and trajectory on a grid with annotated goals, holes, treasures, and teleport portals using Matplotlib.

FALL 2024

Neural Network Modelling using Emergent

University of California, Merced

- Conducted a simulation-based cognitive neuroscience study evaluating the impact of chunking strategies on short-term memory recall using AB-AC paired associate learning tasks.
- Modified a cortical-like neural network to simulate and mitigate catastrophic interference by manipulating context and output layer representations, inspired by chunking theory.
- Performed Euclidean distance analysis to quantify improvements in memory recall and reduction in overlap between memory traces due to chunking.

TEACHING EXPERIENCE

2024 - CURRENT

Graduate Student (Teaching Assistant, Instructor Support)

University of California, Merced

- Serve as a Teaching Assistant for undergraduate-level courses, supporting instruction, grading, and student engagement.
- Responsible for grading assignments, holding office hours, leading lab/discussion sections, and providing individualized support to students.
- Courses supported include Introduction to Human-Computer Interaction and Full Stack Web Development.

PUBLICATIONS & MANUSCRIPTS

Conference Papers:

[C.3] Qhelile Ozias Sibanda, Ahmed Sabbir Arif. 2026. WristWalker: Enhancing Desktop Telepresence Robot Interfaces with Smartwatch-Based Control. Proceedings of the 2026 IEEE Conference on Telepresence (TELEPRESENCE 2026). IEEE. To appear.

[C.2] Tafadzwa Joseph Dube, Qhelile Ozias Sibanda, I. Scott MacKenzie, Ahmed Sabbir Arif. 2026. WinkSwype: A Wink-Based, Hands-Free Shape-Writing Method for VR Using Single-Eye Calibration. Proceedings of the 52nd Graphics Interface Conference (GI 2026). ACM. To appear.

[C.1] Serenity Reign Bassett, Qhelile Ozias Sibanda, Ahmed Sabbir Arif. 2026. Designing for Everyday Mobility: A Market- and User-Driven Study of Adult e-Scooters. International Conference on Pervasive Technologies Related to Assistive Environments (PETRA 2026), CCIS, Springer. To appear.

ACADEMIC SERVICES

2026	Reviewer <i>IEEE Control Systems / Robotics Conference (via PaperCept)</i>
2025 - CURRENT	Co-mentor of Undergraduate Interns <i>University of California, Merced</i>
2025	Student Volunteer <i>Sample Conference, Sample City</i>

PROFESSIONAL AFFILIATIONS

Health Sciences and Research Institute (HSRI) at UC Merced, since Nov 2024

UC Merced Graduate Student Association (GSA) — Delegate, Electrical Engineering & Computer Science, 2024-2025; contributed graphics/design support for GSA communications

Association for Computing Machinery (ACM), member since 2024 (sample entry)

IEEE, member since 2024 (sample entry)

RESEARCH PROJECTS

2026	WinkSwype <i>A Wink-Based, Hands-Free Shape-Writing Method for VR Using Single-Eye Calibration</i> - Co-authored with Tafadzwa Joseph Dube, I. Scott MacKenzie, and Ahmed Sabbir Arif; to appear at the 52nd Graphics Interface Conference (GI 2026). - Designed and evaluated a hands-free shape-writing (swipe-typing) method for VR that uses winks for character selection, calibrated to a single eye.
2024 - CURRENT	WristWalker <i>Smartwatch Gesture vs. Web Control of an Ohmni Telepresence Robot</i> - Built and evaluated a system comparing smartwatch gesture control versus web-based control of an Ohmni telepresence robot. - Ran a 10-participant, IRB-approved, counterbalanced user study; results to appear as "WristWalker: Enhancing Desktop Telepresence Robot Interfaces with Smartwatch-Based Control" at IEEE TELEPRESENCE 2026.

PERSONAL PROJECTS

1	TradeFlow — Trading Journal <i>qtradeflow.com • React</i> Built a professional-grade trading journal replicating and extending TradeZella, supporting multi-account management across forex, stocks, gold, indices, futures, and crypto, with automatic P&L calculations, analytics dashboards, and playbook/goal tracking.
2	MQL5/MQL4 Trading Tools <i>MetaTrader 4/5, Deriv</i> Built a suite of automated trading tools for MetaTrader covering forex pairs, synthetic indices, gold, and crypto, including EMA crossover expert advisors, a supply and demand zone indicator, and an ICC (Indication-Correction-Continuation) state-machine expert advisor.
3	3D Race Game <i>OpenGL, Unity3D, C programming</i> Developed this game based on the retro console brick game, using data structure algorithms involving stacks and queues. The game was then ported into virtual reality.

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Cartoon Character Creation

Blender

- Modeled a functional 3D human cartoon character from scratch in Blender, covering the full pipeline from base mesh sculpting to a rigged, posable character.
 - Applied topology, UV unwrapping, and texturing/shading techniques to give the character a clean, stylized cartoon look.
 - Set up basic rigging and armature for the character to support posing and animation.
- 5

WhatsApp Sports Betting Platform

Machine learning algorithms, Python • Final year project (BSc Honors)

- Developed an interactive chatbot that took user input from WhatsApp and processed bets placed based on given odds and stats of the game. The chatbot was linked to a database management system for user management.
- 6

CSE 005 Teaching Materials

Markdown, HTML, Excel/VBA

- Created answer keys, grading guides, and interactive Kahoot-style classroom review games for an introductory computer science course, covering web fundamentals, software concepts, and data flow diagrams.

POSTERS & PRESENTATIONS

Presentations:

[Pr.2] "WinkSwype: A Wink-Based, Hands-Free Shape-Writing Method for VR Using Single-Eye Calibration," 52nd Graphics Interface Conference (GI 2026).

[Pr.1] "Designing for Everyday Mobility: A Market- and User-Driven Study of Adult e-Scooters," International Conference on Pervasive Technologies Related to Assistive Environments (PETRA 2026).

PROFESSIONAL EXPERIENCE

- 2021 - 2024

Parttime Programmer, Spacebar Holdings

Bulawayo, Zimbabwe

- Helped build Town Council systems (Housing system, Equipment Management System, Town Planning system).
 - Upgraded the old town council housing system from a desktop system to a web-based system.
 - Responsible for user training and system maintenance.
 - Worked on a virtual reality-based town house tour.
- 2020 - 2022

Financial Analyst, Bullex Investments

Houston, TX • under Brian Fontenot

- Analyzed markets and supported investment decision-making as part of the team led by Brian Fontenot.
 - Contributed to financial reporting and research supporting the firm's investment strategy.
- 2020 - 2021

System Developer, ForexEmpireKings

Bulawayo, Zimbabwe

- Helped build trading indicators for the forex market using MQL4 and MQL5 programming languages.
 - Debugged code for trading expert advisors and indicators.
- 2019 - 2021

Systems Analyst & Developer (Intern), Hansa World

Zimbabwe

- Tested HansaWorld Standard ERP System for bugs, evaluated system operating speed, and documented findings.
 - Responsible for creating system builds for new clients; conducted user training, system testing, and system implementation for clients.
 - Programmed custom Point of Sale modules for Standard Hansa World ERP using their custom programming language, HAL.
- 2016 - 2017

Data Analyst, Cash Crusaders

South Africa

- Responsible for checking merchandise and packaging of goods.
- Responsible for managing daily business reports of sales such as aging stock and bestselling products.

SKILLS

Programming Languages:

Python, Java, C, C++, C#, SQL, Bash, GO

Web Development:

HTML5, PHP, JavaScript, CSS

Applications / IDEs:

Unity3D, PyCharm, Android Studio, IntelliJ, Eclipse, Visual Studio, Jupyter Notebook, GitHub, GDB, Valgrind, SQLite, Bison/Yacc

Machine Learning:

Chatbot Creation, Data Modelling, Google TensorFlow, Bayes Nets for Probabilistic Prediction

Personal:

Excellent communication skills, able to work and collaborate in a team, able to work under pressure

FELLOWSHIPS & AWARDS

Best Student Award

Best Actor Award

Maths Wizard Award

Smartest Boy Award

Forex Funded Account Trader

Best Worker of the Month | IBS

HOBBIES

Reading • Socializing • Piano Playing • VR Gaming

REFERENCES

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